

[5] LT3042 - LDO

6V to 5V Analog

TYPICAL APPLICATIONS
Low Noise Wheatstone Bridge Power Supply

RESISTOR TOLERANCE	BRIDGE PSRR	NOISE AT Vbridge USING LT3042	NOISE AT Vbridge USING LT3042
PERFECT MATCHING	INFINITE	—	—
1%	40dB	200nVrms	8µVrms
5%	26dB	1000nVrms	42.5µVrms

IN (pins 1-2): LT3042 power input (6V), to be decoupled with CIN.

EN/UV (pin 3): enables/disables the regulator and allows an UVLO threshold (often tied to IN for "always ON").

PG (pin 4): open-collector Power-Good output (requires a pull-up resistor).

ILIM (pin 5): current limit programming/monitoring (often tied to GND if unused).

PGFB (pin 6): input that sets the Power-Good threshold via a resistor divider from VOUT.

SET (pin 7): sets the output voltage via RSET (and CSET reduces noise / provides soft-start).

GND (pin 8): circuit ground reference.

OUTS (pin 9): Kelvin output sense (to be connected to the + of COUT / load point).

OUT (pin 10): regulated output (5V), to be decoupled with COUT.

EPAD (exposed pad): ground + thermal dissipation (to be soldered to the GND plane).

DESIGN NOTE:

Ref Condo for SET PIN (important pour les vibrations P16)
KYOCERA AVX TAJA225K016RNJ (2.2 µF, 16 VDC, condensateur tantale solide SMD moulé, 1206 / 3216 (A case), ±10%, ESR 6.5 Ω).

$V_{OUT} \approx I_{SET} \times R_{SET}$
 $I_{SET} \approx 100 \mu A$

Donc :
 $R_{SET} \approx V_{OUT} / I_{SET}$
 $R_{SET} \approx 5.0 V / 100 \mu A \approx 50 k\Omega$
(=> valeur standard recommandée : 49.9 kΩ)

$V_{OUT}(PG_THRESHOLD) \approx 0.3 V \times (1 + R_{PG2}/R_{PG1})$
 $PGFB_THRESHOLD \approx 0.3 V$

Donc :
 $R_{PG2} \approx R_{PG1} \times (V_{OUT}(PG)/0.3 V - 1)$

Exemple (PG OK à ~4.8 V) :
Choix $R_{PG1} = 49.9 k\Omega$
 $R_{PG2} \approx 49.9 k\Omega \times (4.8/0.3 - 1) \approx 748.5 k\Omega$
(=> valeur standard recommandée : 750 kΩ)

Design guide line

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Convertisseur 5VA

About

The LT®3042 is a high performance low dropout linear regulator featuring LTC's ultralow noise and ultrahigh PSRR architecture for powering noise sensitive RF applications. Designed as a precision current reference followed by a high performance voltage buffer, the LT3042 can be easily paralleled to further reduce noise, increase output current and spread heat on the PCB. The device supplies 200mA at a typical 350mV dropout voltage. Operating quiescent current is nominally 2mA and drops to <<1µA in shutdown. The LT3042's wide output voltage range (0V to 15V) while maintaining unitygain operation provides virtually constant output noise, PSRR, bandwidth and load regulation, independent of the programmed output voltage. Additionally, the regulator features programmable current limit, fast start-up capability and programmable power good to indicate output voltage regulation. The LT3042 is stable with a minimum 4.7µF ceramic output capacitor. Built-in protection includes reverse battery protection, reverse current protection, internal current limit with foldback and thermal limit with hysteresis. The LT3042 is available in thermally enhanced 10-Lead MSOP and 3mm × 3mm DFN package

TEST POINT

	Comments:		Company:		Variant:	Git Hash:	
	LT3042 Analog Devices		OPEN THRUST LAB				b3867f9
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			AD7190		OPEN_WEIGHT		
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/Project Architecture/LT3042_6V_to_5VA/					A3	5 of 10	